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To cite this article: Yuji Ohishi *et al* 2016 *Jpn. J. Appl. Phys.* **55** 051301

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Thermoelectric properties of gallium-doped p-type germanium

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Received November 3, 2015; accepted February 13, 2016; published online March 30, 2016

In this study, the temperature-dependent thermoelectric properties of p-type single-crystal Ge, which is a useful material for thermoelectric applications owing to its significantly high carrier mobility, were investigated. The thermoelectric properties of Ga-doped (5.7×10^{16} , 3.4×10^{18} , and $1.0 \times 10^{19} \text{ cm}^{-3}$) p-type single-crystal Ge were measured from room temperature to 770 K. The sample with a carrier concentration of $1.0 \times 10^{19} \text{ cm}^{-3}$ showed the highest thermoelectric figure of merit, ZT , over the entire measured temperature range. The maximum ZT value was 0.06 at 650 K. A theoretical model based on the Boltzmann transport equation with relaxation-time approximation was developed and quantitatively reproduced the experimentally observed data. The optimal impurity concentration predicted by this model was $3 \times 10^{19} \text{ cm}^{-3}$ at 300 K and increased with temperature. © 2016 The Japan Society of Applied Physics

1. Introduction

Owing to its high carrier mobility, germanium (Ge) is expected to replace silicon (Si) as a channel material in high-performance metal–oxide–semiconductor field-effect transistors.¹⁾ The carrier mobilities (μ) of electrons ($3900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) and holes ($1900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) in lightly doped Ge are respectively two and four times higher than those in Si.²⁾ The high carrier mobility of Ge is also advantageous for thermoelectric (TE) applications.

TE generators directly convert heat into electrical energy and have been attracting much attention in waste heat recovery.^{3,4)} The efficiency of TE conversion depends on the material's figure of merit $ZT = (S^2 \sigma / \kappa) T$, where S is the Seebeck coefficient, σ is the electrical conductivity, κ is the total thermal conductivity (sum of the electrical thermal conductivity κ_e and lattice thermal conductivity κ_l), and T is the absolute temperature.⁵⁾ σ is expressed as $ne\mu$, where n is the carrier concentration and e is the charge of an electron. Thus, high carrier mobility μ results in high σ and high ZT . The TE properties of Si-based materials have been studied intensively.^{6–9)} In contrast, although there are several studies that individually focus on the physical properties of Ge such as electrical conductivity,¹⁰⁾ the Seebeck coefficient,¹¹⁾ and thermal conductivity,^{12,13)} the TE properties of Ge have not been discussed yet.

The most appropriate approach to understanding the basic physical properties of materials is to measure the properties using single crystals. Recently, we have measured and reported the basic TE properties of single-crystal Si.¹⁴⁾ Similarly, this paper is intended to present the TE properties of Ge, based on results found by studying Ga-doped (5.7×10^{16} , 3.4×10^{18} , and $1.0 \times 10^{19} \text{ cm}^{-3}$) p-type Ge. The Ge used in this study was in the form of single-crystal wafers. The samples were studied at temperatures ranging from room temperature to 770 K. Using the experimentally observed data, we developed theoretical models for estimating the temperature dependence of the TE properties of bulk Ge with various doping concentrations.

2. Experimental procedure

Commercially available single-crystal Ge wafers were used in

Table I. Information on the samples used in this study. The samples were cut from Ge(100) wafers.

Sample code	Thickness (mm)	Dopant	Carrier density (cm^{-3})	Resistivity ($\Omega \text{ cm}$)
P1	0.35	Ga	1.0×10^{19}	7.8×10^{-2}
P2	0.38	Ga	3.4×10^{18}	5.7×10^{-3}
P3	0.54	Ga	5.7×10^{16}	2.8×10^{-3}

this study. Charge carrier concentrations and mobilities were determined by Hall measurements at a magnetic field of 0.5 T in a van der Pauw geometry with a Toyo ResiTest 8340 system. The measured carrier density and resistivity values of the samples are shown in Table I. The electrical conductivity was measured using a four-point current-switching technique, and the Seebeck coefficient was measured by a static DC method using commercial equipment (Ulvac ZEM-3). A bar-type sample of 2.8–4.5 mm width, 0.36–0.53 mm height, and 10 mm length was cut from the Ge wafer and used for these measurements. The thermal conductivity was calculated from the heat capacity,¹⁵⁾ thermal diffusivity, and density. The thermal diffusivity was measured by a laser flash technique using Netzsch LFA 467 from room temperature to 773 K. Ge wafers with (100) crystal orientation were used in the experiment, as noted in Table I. The thermal diffusivity was measured parallel to the (100) direction, and the electrical properties were measured perpendicular to the (100) direction.

3. Model for thermoelectric transport in Ge

The electrical conductivity and Seebeck coefficient of Ge were calculated using the Boltzmann transport equation with the relaxation-time approximation.¹⁶⁾ The model used is similar to those developed by Minnich et al.¹⁷⁾ and Lee et al.¹⁸⁾ Since electrons are scattered mainly by ionized impurities and acoustic phonons in crystalline Ge, only these two scattering mechanisms were considered in this theoretical model. The functional forms were taken from Ref. 19. The model used in this study assumes four-band transport involving a light hole band, a heavy hole band, a split-off hole band for the valence band, and one band at an L point for the conduction band. This conduction L band was

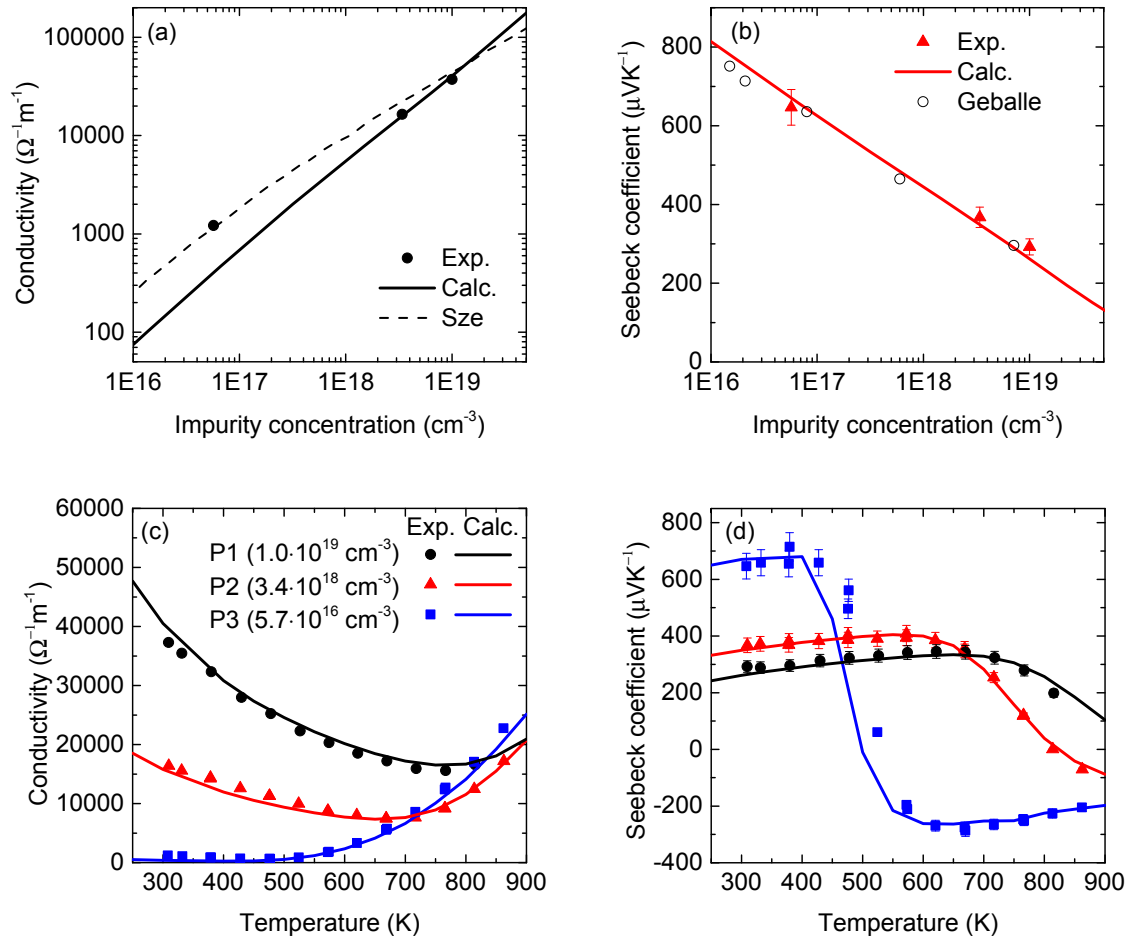


Fig. 1. (Color online) (a) Electrical conductivity and (b) Seebeck coefficient of Ge as a function of impurity concentration at room temperature. Experimental data of single-crystal Ge reported by Sze and Jvin¹⁰⁾ and Geballe and Hull¹¹⁾ are plotted together. (c, d) Temperature dependences of experimental (filled symbols) and calculated (curves) electrical conductivities and Seebeck coefficients, respectively.

introduced to reproduce the behavior of the intrinsic region at high temperatures. The valence band edges were approximated by parabolic bands and the conduction band edge was approximated by a nonparabolic band. Since we have reported the TE properties of Si using a very similar approach,¹⁴⁾ the details of the theoretical model, such as functional forms and parameters, are not shown here, but they can be found in the online supplementary data at <http://stacks.iop.org/JJAP/55/051301/mmedia>.

To calculate the lattice thermal conductivity, we used the model proposed by Morelli et al.,²⁰⁾ which can quantitatively calculate the thermal conductivity of nondoped Ge. This model is also based on the Boltzmann transport equation with the relaxation-time approximation. The phonon scattering mechanisms considered in this model include phonon-isotope scattering, phonon-phonon umklapp scattering, and phonon-phonon normal scattering, all of which are characterized by the relaxation time. Since only a slight difference was observed between the samples, we ignored the impurity concentration dependence of the thermal conductivity. This assumption is reasonable since the phonon-impurity scattering rate increases with the differences in mass and radius between the substituted atoms and the matrix.²¹⁾ The impurity scattering rate due to Ga is low owing to the similar ionic radii and atomic masses of Ga and Ge. The parameters used in this calculation are exactly the same as those indicated in Ref. 20 and are therefore not reproduced here.

4. Results and discussion

4.1 Carrier transport property

Assuming the complete ionization of the dopants, the carrier density is considered to be equal to the impurity concentration. Figures 1(a) and 1(b) show the measured electrical conductivity and Seebeck coefficient, respectively, at room temperature as a function of impurity concentration. Data from Refs. 10 and 11 are plotted together, showing that the data observed in this study are in agreement with the literature data.

The temperature dependences of the electrical conductivity and Seebeck coefficient are shown in Figs. 1(c) and 1(d), respectively. These figures show that the model quantitatively reproduces the temperature dependences of the electrical conductivity and Seebeck coefficient over the entire temperature range for all the samples. Figure 1(c) shows that, with increasing temperature, the electrical conductivity is first reduced by electron-phonon scattering and then starts to increase owing to the thermal excitation of carriers, which causes the rapid reduction in Seebeck coefficient as shown in Fig. 1(d).

The Seebeck coefficient of sample P3 changes from positive to negative at around 550 K, which suggests that the main carriers in sample P3 change from holes to electrons above 550 K. This change can be attributed to the difference in carrier mobility between electrons and holes in Ge. In the intrinsic region, not only holes but also thermally excited

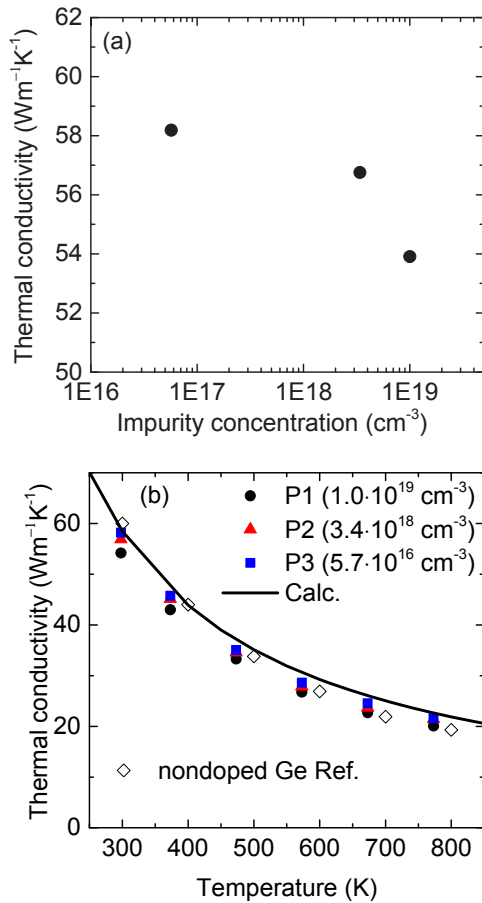


Fig. 2. (Color online) (a) Thermal conductivity as a function of impurity concentration. (b) Experimental (filled symbols) and calculated (curves) temperature-dependent thermal conductivities of Ge.

electrons contribute to the electrical transport. Since the carrier mobility of electrons is much higher than that of holes in Ge, electrons become dominant in electrical transport in sample P3 in the intrinsic temperature region.

4.2 Thermal transport properties

Figures 2(a) and 2(b) show the observed thermal conductivities in this study. The thermal conductivity is a result of lattice and the electronic contributions, and is related to the electrical conductivity by the Wiedemann–Franz law. On the basis of this law, the electronic thermal conductivity is calculated as $\kappa_e = LT\sigma$, where L is the Lorenz factor ($2.44 \times 10^{-8} \text{ J}^2 \text{ K}^{-2} \text{ C}^{-2}$). The electronic thermal conductivity was calculated from the measured electrical conductivity and was found to be less than $0.8 \text{ W m}^{-1} \text{ K}^{-1}$ for all the samples over the entire temperature range of the measurements. Because this value is smaller than the lattice thermal conductivity, we considered the electrical thermal conductivity to be negligible and the phonon thermal conductivity to be dominant for all the samples.

The impurity concentration dependence of thermal conductivity at room temperature is shown in Fig. 2(a). The thermal conductivities of the Ga-doped Ge ($58\text{--}54 \text{ W m}^{-1} \text{ K}^{-1}$) at room temperature are slightly lower than those of the nondoped single-crystal Ge ($60 \text{ W m}^{-1} \text{ K}^{-1}$).^{12,13} The thermal conductivity decreases with increasing carrier concentration possibly because of phonon–impurity scattering and phonon–carrier scattering.

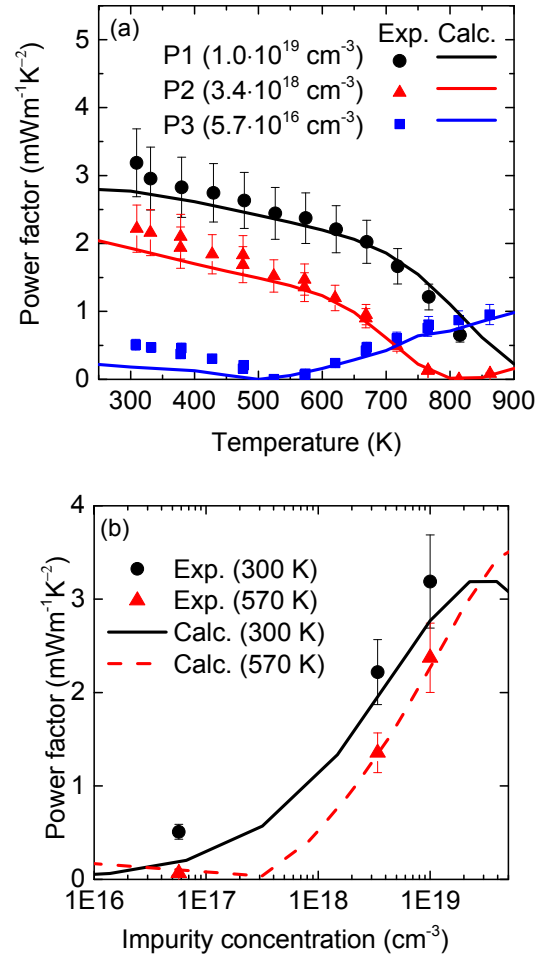


Fig. 3. (Color online) (a) Experimental (filled symbols) and calculated (curves) temperature-dependent power factors of Ge. (b) Power factor of Ge as a function of impurity concentration from 300 to 570 K. The symbols and curves represent the experimental and calculated data, respectively.

Figure 2(b) shows the temperature dependence of the thermal conductivity. The thermal conductivity decreases with increasing temperature owing to the phonon–phonon scattering. The variation in the thermal conductivity of the samples is small, and the calculated values are in agreement with the experimentally obtained values. This is because phonon–impurity scattering is not important for the Ga-doped Ge samples owing to the similar ionic radii and atomic masses of Ga and Ge as mentioned in the theoretical model section. The phonon–electron scattering is also not important since the carrier concentrations in the Ge samples used in this study are relatively low (below $1.0 \times 10^{19} \text{ cm}^{-3}$).

4.3 Thermoelectric properties

Figures 3(a) and 3(b) present the power factor as a function of temperature and impurity concentration, respectively. The model successfully represents the experimentally obtained power factor values. The power factor decreases with increasing temperature because of the reduction in electrical conductivity. The power factor of sample P3 starts to increase above 550 K because of the thermally excited electrons. Figure 3(b) shows that the experimentally observed power factor increases with increasing impurity concentration. The model predicts that the maximum power factor will

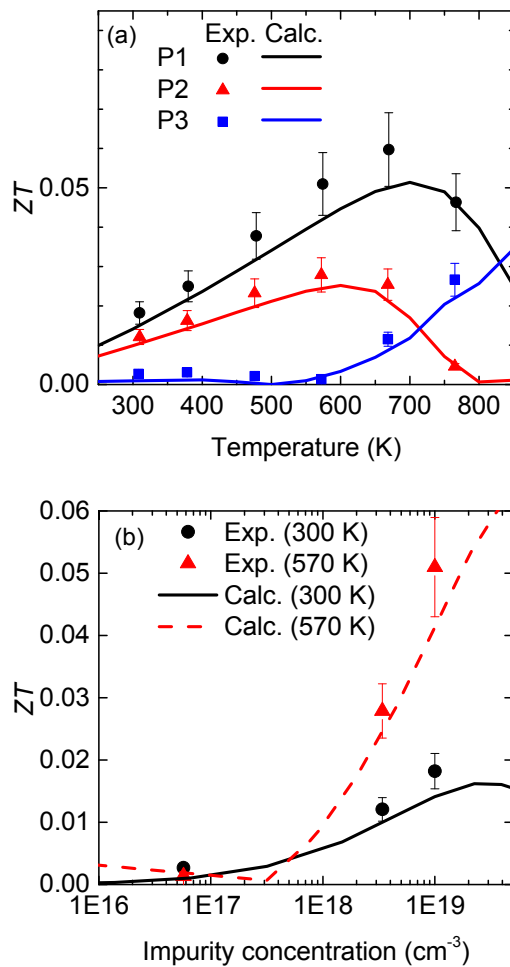


Fig. 4. (Color online) ZT of Ge as a function of (a) temperature and (b) impurity concentration. The symbols and curves represent the experimental and calculated data, respectively.

be obtained at a carrier concentration of $3 \times 10^{19} \text{ cm}^{-3}$ at 300 K and will decrease with temperature.

The temperature and impurity concentration dependences of ZT are shown in Figs. 4(a) and 4(b), respectively. It was found that the calculated results were in excellent agreement with the experimentally observed data. Figure 4(a) shows that ZT first increases with temperature, and starts to decrease owing to the thermal excitation of electrons. Since the thermally excited electrons become the main carriers in sample P3 above 550 K, a steep increase in ZT was observed for sample P3 in this temperature range. As in the case of the power factor, ZT increases with increasing impurity concentration. The model predicts that the maximum ZT will be obtained at a carrier concentration of $3 \times 10^{19} \text{ cm}^{-3}$ at 300 K and will increase with temperature.

5. Conclusions

In this study, the temperature-dependent TE properties of Ga-doped (5.7×10^{16} , 3.4×10^{18} , $1.0 \times 10^{19} \text{ cm}^{-3}$) p-type single-crystal Ge were measured from room temperature to 770 K and a theoretical model was developed to predict the TE properties of Ge using the Boltzmann transport equation. The calculated values were in good agreement with the experimentally observed data, indicating that our model can be used to predict the TE properties of Ge. ZT increases with the impurity concentration in the measured impurity concentration range. The model predicted that the maximum ZT will be obtained at a carrier concentration of $3 \times 10^{19} \text{ cm}^{-3}$ at 300 K and will increase with temperature. We hope that the experimentally observed data and the developed theoretical model will help develop Ge-based TE materials.

Acknowledgments

This work was supported by the Advanced Low Carbon Technology Research and Development Program (ALCA) of the Japan Science and Technology Agency. We wish to acknowledge the assistance of Netzsch Japan in obtaining the thermal diffusivity data.

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